ΔΙΚΤΥΑ ΥΠΟΛΟΓΙΣΤΩΝ (2η Εργαστηριακή Άσκηση)

ΕΠΩΝΗΜΟ:ΚΙΟΥΡΤ ΜΕΧΜΕΤΑΛΗ

ΟΝΟΜΑ:ΑΧΜΕΤ

ΑΜ:1084672

Ερωτήματα

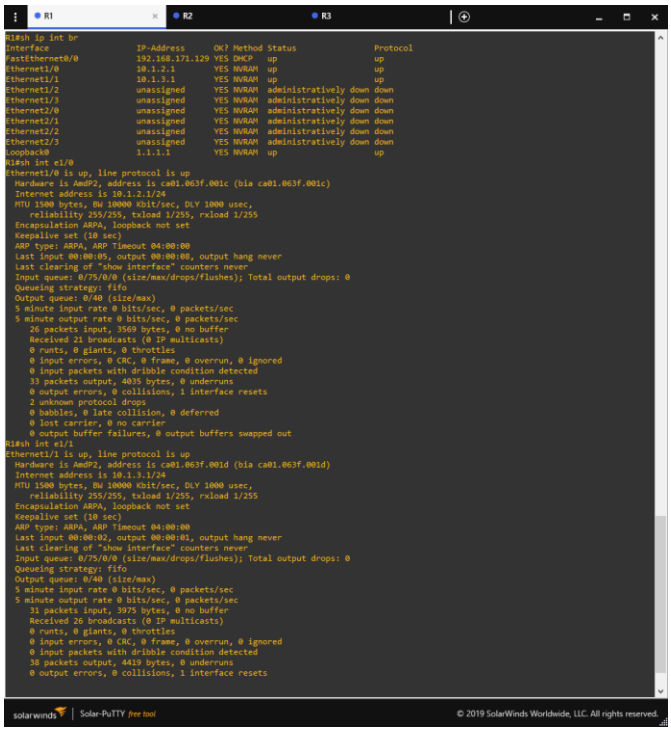
1) Αρχικά για την πρώτη ερώτηση παρατηρούμε ότι με τις εντολές sh ip int br έχουμε αναθέσει σε κάθε δρομολογητή τα IP με έναν τρόπο που να μπορούνε να επικοινωνούν περεταίρω και με sh int <όνομα διεπαφής> παρατηρούμε ότι υπάρχουν οι συνδέσεις e1/0 και e1/1 για να μπορούμε να συνεχίσουμε την διαδικασία επικοινωνίας των δρομολογητών.

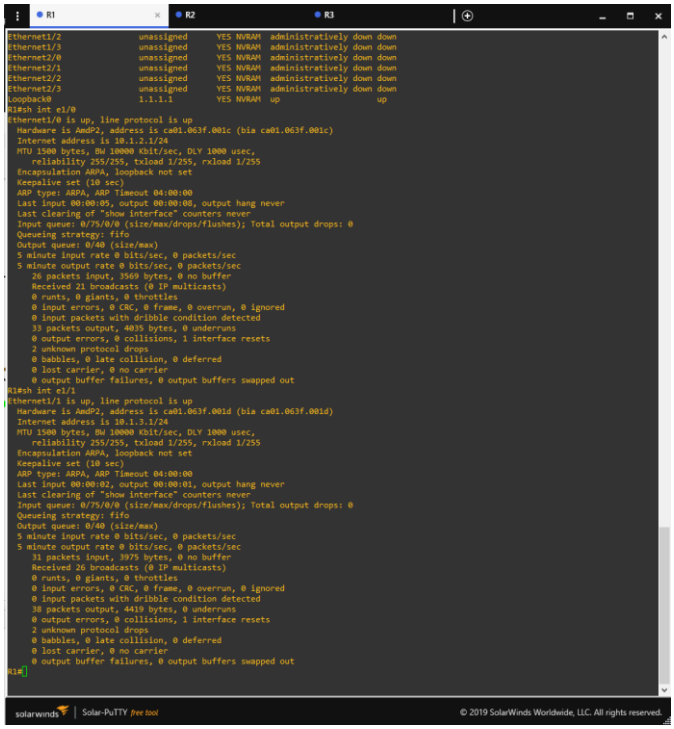
Τα IP που έχουμε βάλει είναι :

	e1/0	e1/1
R1	10.1.2.1	10.1.3.1
R2	10.1.2.2	10.1.4.1
R3	10.1.3.2	10.1.4.2

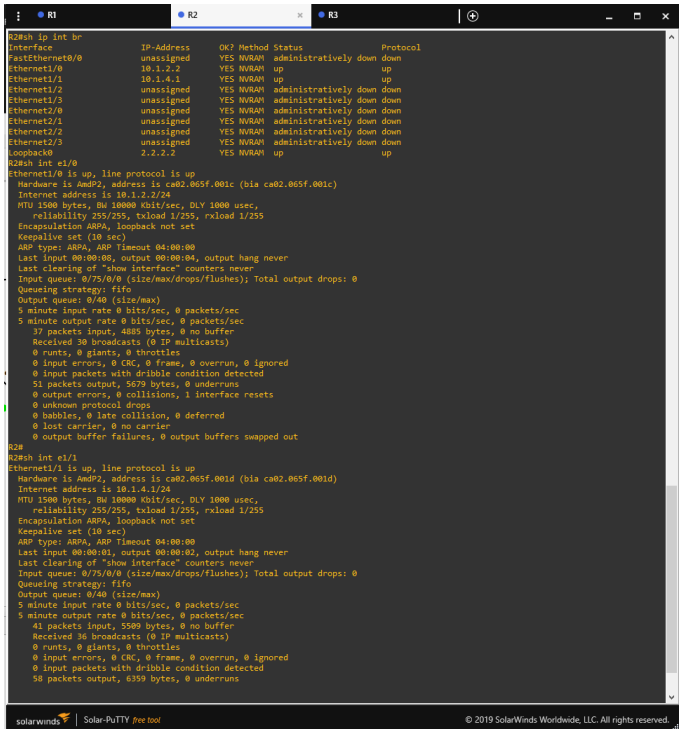
Από τα χρώματα μπορούμε να βεβαιωθούμε ποια IP του δρομολογητή επικοινωνεί με ποιον δρομολογητή. Με αποτέλεσμα να έχουμε μία σωστή δικτύωση για την συνέχεια τις εργασίας

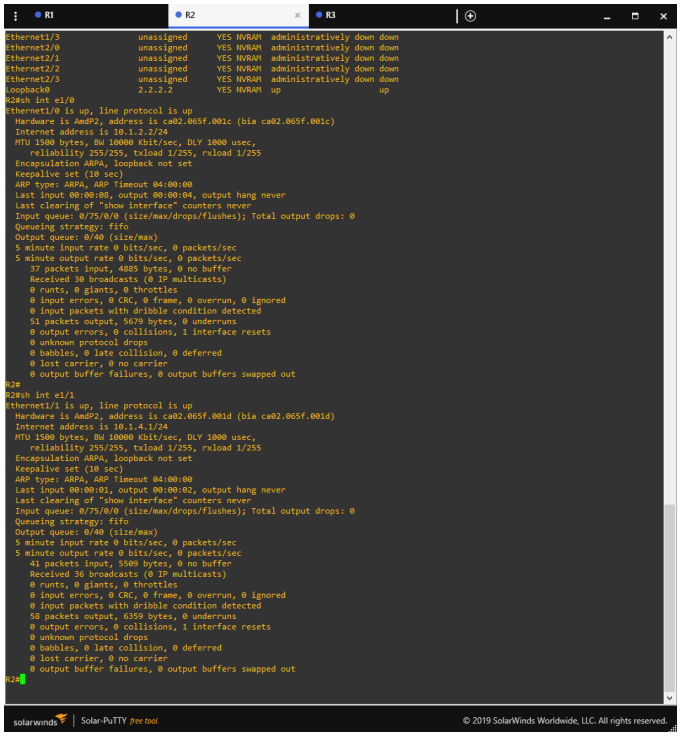
Για το R1:



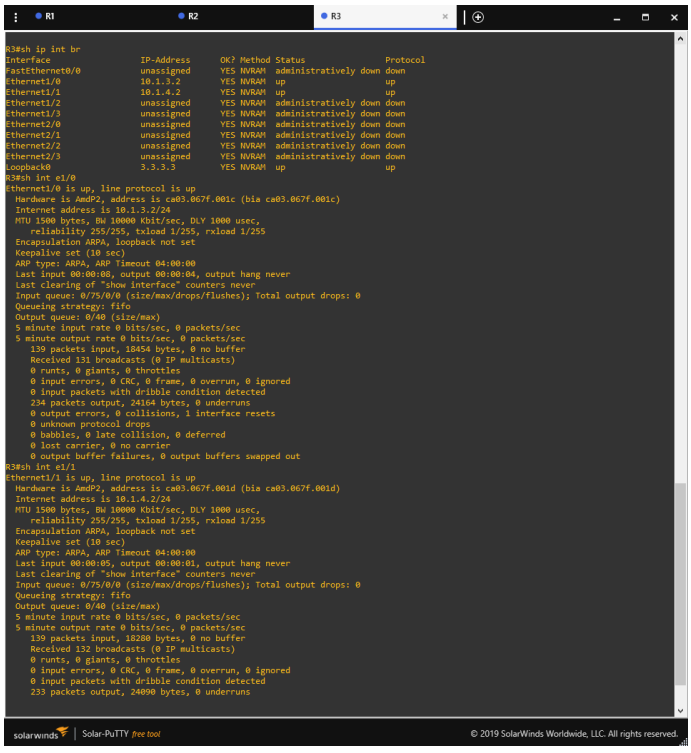

R1
R2
R3
Ethernet1/2 unassigned YES NVRAM administratively down down
Ethernet1/3 unassigned YES NVRAM administratively down down
Ethernet2/0 unassigned YES NVRAM administratively down down
Ethernet2/1 unassigned YES NVRAM administratively down down
Ethernet2/2 unassigned YES NVRAM administratively down down
Ethernet2/3 unassigned YES NVRAM administratively down down
Loopback0 1.1.1.1 YES NVRAM up up
R1#sh int e1/0
Ethernet1/0 is up, line protocol is up
Hardware is AmdP2, address is ca01.063f.001c (bia ca01.063f.001c)
Internet address is 10.1.2.1/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:05, output 00:00:08, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
26 packets input, 3569 bytes, 0 no buffer
Received 21 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
33 packets output, 4035 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
2 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R1#sh int e1/1
Ethernet1/1 is up, line protocol is up
Hardware is AmdP2, address is ca01.063f.001d (bia ca01.063f.001d)
Internet address is 10.1.3.1/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:02, output 00:00:01, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
31 packets input, 3975 bytes, 0 no buffer
Received 26 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
38 packets output, 4419 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
2 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R1#
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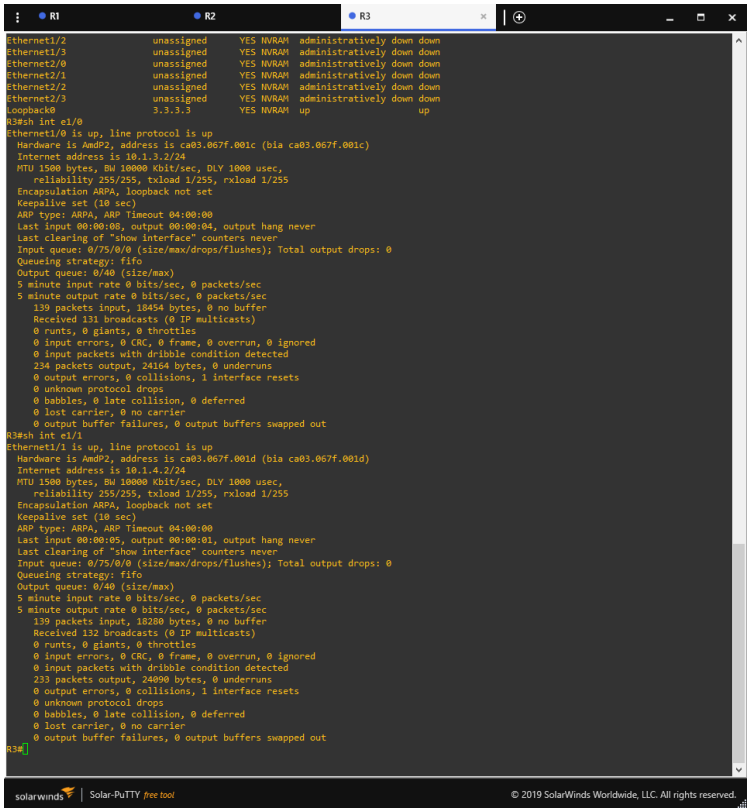
Για το R2:


R1
R2
R3
R2#sh ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES NVRAM administratively down down
Ethernet1/0 10.1.2.2 YES NVRAM up up
Ethernet1/1 10.1.4.1 YES NVRAM up up
Ethernet1/2 unassigned YES NVRAM administratively down down
Ethernet1/3 unassigned YES NVRAM administratively down down
Ethernet2/0 unassigned YES NVRAM administratively down down
Ethernet2/1 unassigned YES NVRAM administratively down down
Ethernet2/2 unassigned YES NVRAM administratively down down
Ethernet2/3 unassigned YES NVRAM administratively down down
Loopback0 2.2.2.2 YES NVRAM up up
R2#sh int e1/0
Ethernet1/0 is up, line protocol is up
Hardware is AmdP2, address is ca02.065f.001c (bia ca02.065f.001c)
Internet address is 10.1.2.2/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:00, output 00:00:04, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
37 packets input, 4885 bytes, 0 no buffer
Received 30 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
51 packets output, 5679 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R2#
R2#sh int e1/1
Ethernet1/1 is up, line protocol is up
Hardware is AmdP2, address is ca02.065f.001d (bia ca02.065f.001d)
Internet address is 10.1.4.1/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:02, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
41 packets input, 5509 bytes, 0 no buffer
Received 36 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
58 packets output, 6359 bytes, 0 underruns
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R1
R2
R3
Ethernet1/3 unassigned YES NVRAM administratively down down
Ethernet1/0 unassigned YES NVRAM administratively down down
Ethernet2/1 unassigned YES NVRAM administratively down down
Ethernet2/2 unassigned YES NVRAM administratively down down
Ethernet2/3 unassigned YES NVRAM administratively down down
Loopback0 2.2.2.2 YES NVRAM up up
R2#sh int e1/0
Ethernet1/0 is up, line protocol is up
Hardware is AmdP2, address is ca02.065f.001c (bia ca02.065f.001c)
Internet address is 10.1.2.2/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:08, output 00:00:04, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
37 packets input, 4885 bytes, 0 no buffer
Received 30 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
51 packets output, 5679 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R2#
R2#sh int e1/1
Ethernet1/1 is up, line protocol is up
Hardware is AmdP2, address is ca02.065f.001d (bia ca02.065f.001d)
Internet address is 10.1.4.1/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:02, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
41 packets input, 5509 bytes, 0 no buffer
Received 36 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
58 packets output, 6359 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R2#
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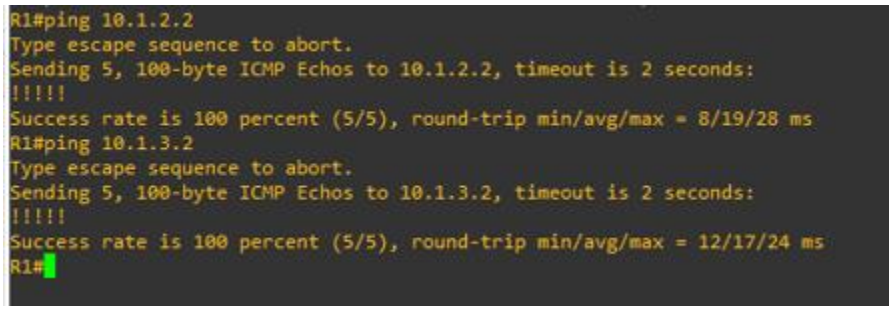
Για το R3:


R1
R2
R3
R3#sh ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES NVRAM administratively down down
Ethernet1/0 10.1.3.2 YES NVRAM up up
Ethernet1/1 10.1.4.2 YES NVRAM up up
Ethernet1/2 unassigned YES NVRAM administratively down down
Ethernet1/3 unassigned YES NVRAM administratively down down
Ethernet2/0 unassigned YES NVRAM administratively down down
Ethernet2/1 unassigned YES NVRAM administratively down down
Ethernet2/2 unassigned YES NVRAM administratively down down
Ethernet2/3 unassigned YES NVRAM administratively down down
Loopback0 3.3.3.3 YES NVRAM up up
R3#sh int e1/0
Ethernet1/0 is up, line protocol is up
Hardware is AmdP2, address is ca03.067f.001c (bia ca03.067f.001c)
Internet address is 10.1.3.2/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:08, output 00:00:04, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
139 packets input, 18454 bytes, 0 no buffer
Received 131 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
234 packets output, 24164 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R3#sh int e1/1
Ethernet1/1 is up, line protocol is up
Hardware is AmdP2, address is ca03.067f.001d (bia ca03.067f.001d)
Internet address is 10.1.4.2/24
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:05, output 00:00:01, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
139 packets input, 18280 bytes, 0 no buffer
Received 132 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
233 packets output, 24090 bytes, 0 underruns
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2) Στην δεύτερη ερώτηση θα εκτελέσουμε την εντολή ping για να έχουμε μία σωστή επικοινωνία των δρομολογητών αλλά και να μπορούμε να βεβαιωθούμε ότι έχουμε κάνει σωστά την ανάθεση.

Ping το R1 για τους δρομολογητές R2 και R3:



Ping το R2 για τους δρομολογητές R1 και R3:



Ping το R3 για τους δρομολογητές R1 και R2:

```
R3#ping 10.1.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/28 ms
R3#ping 10.1.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/24 ms
R3#
```

3) Με την εντολή ospf γράφουμε το LoopBack του δρομολογητή και αναθέτουμε ένα default δίκτυο που μπορούμε στην συνέχεια να παρατηρήσουμε με το command **sh ip ospf neigh** βλέπουμε τα δίκτυα που έχουμε κάνει ping.

Για το R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 0.0.0.0 255.255.255.255 area 0
R1(config-router)#end
R1#
*Mar 13 13:34:42.187: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Mar 13 13:35:55.787: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet1/0 from LOADING to FULL, Loading Done
R1#
*Mar 13 13:36:39.571: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet1/1 from LOADING to FULL, Loading Done
R1#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:32    10.1.3.2        Ethernet1/1
2.2.2.2           1   FULL/BDR        00:00:35    10.1.2.2        Ethernet1/0
R1#
```

Για το R2:

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2 255.255.255.255 area 0
                                    ^
% Invalid input detected at '^' marker.

R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 0.0.0.0 255.255.255.255 area 0
R2(config-router)#end
R2#
*Mar 13 13:35:55.815: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Ethernet1/0 from LOADING to FULL, Loading Done
R2#
*Mar 13 13:35:57.419: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Mar 13 13:36:39.575: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet1/1 from LOADING to FULL, Loading Done
R2#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:36    10.1.4.2        Ethernet1/1
1.1.1.1           1   FULL/DR         00:00:38    10.1.2.1        Ethernet1/0
R2#
```

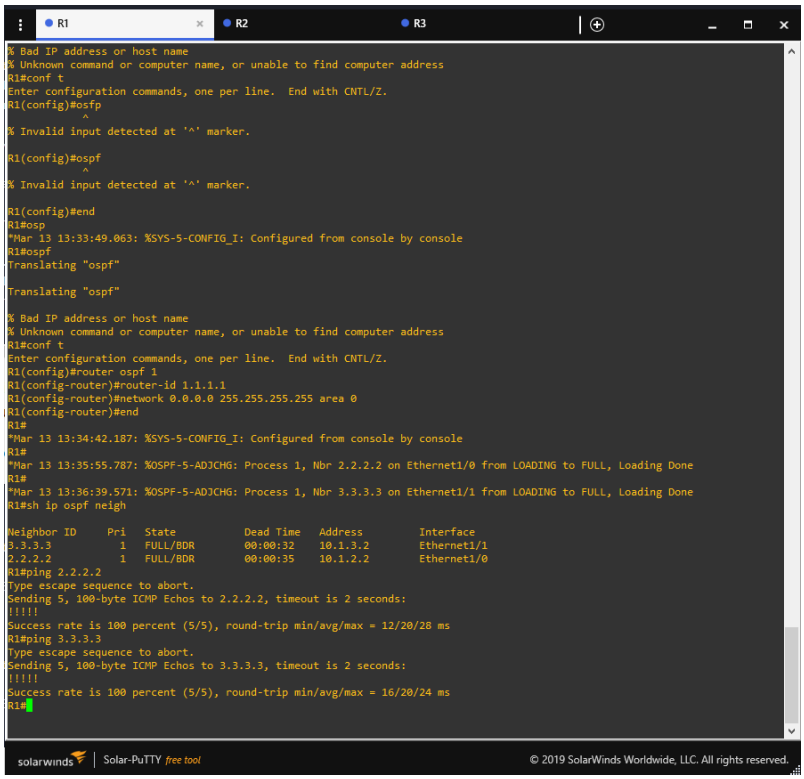
Για το R3:

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 0.0.0.0 255.255.255.255 area 0
R3(config-router)#
*Mar 13 13:36:39.595: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet1/1 from LOADING to FULL, Loading Done
*Mar 13 13:36:39.599: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Ethernet1/0 from LOADING to FULL, Loading Done
R3(config-router)#end
R3#
*Mar 13 13:36:52.115: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip ospf neigh

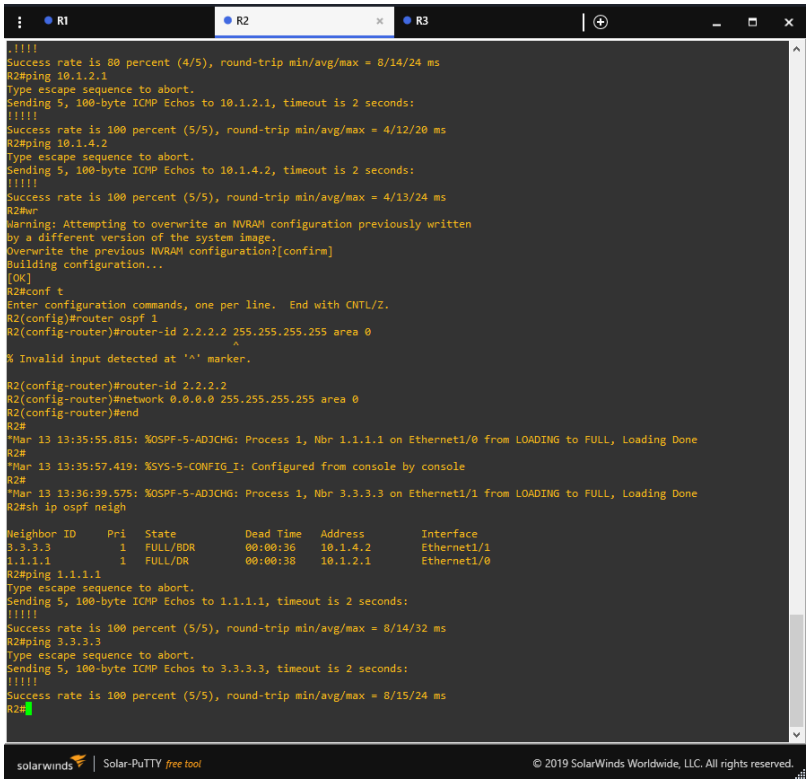
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    10.1.4.1        Ethernet1/1
1.1.1.1           1   FULL/DR         00:00:34    10.1.3.1        Ethernet1/0
R3#
```

4) Ξανά χρησιμοποιούμε την εντολή ping και για να βεβαιωθούμε ότι βλέπουν οι δρομολογητές τις διεπαφές και υπάρχει η σύνδεση και επικοινωνία μεταξύ των LoopBack.

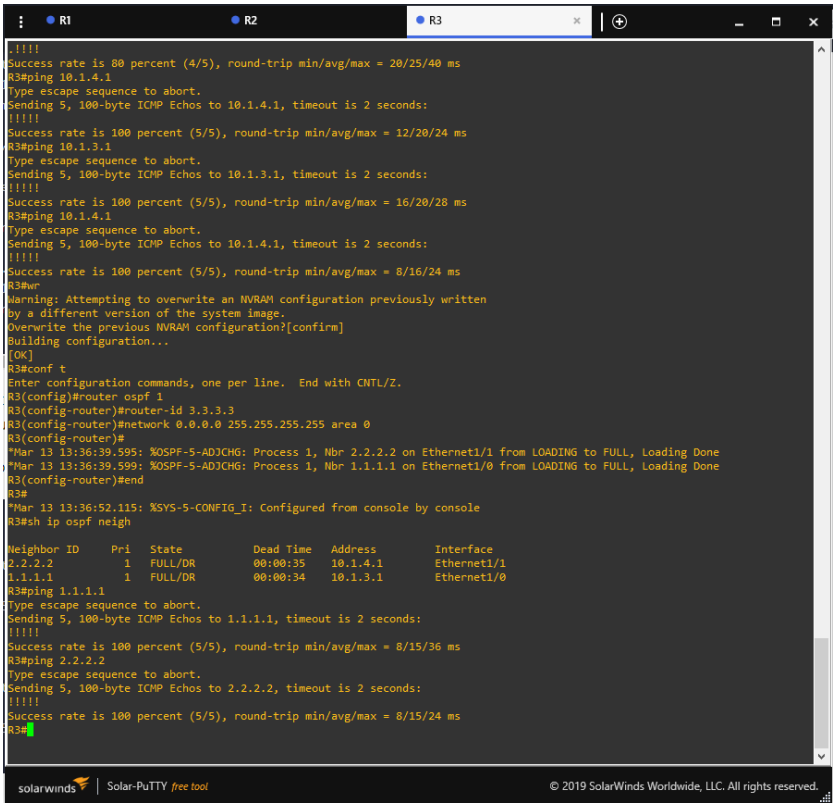
Για τον δρομολογητή R1:



Για τον δρομολογητή R2:

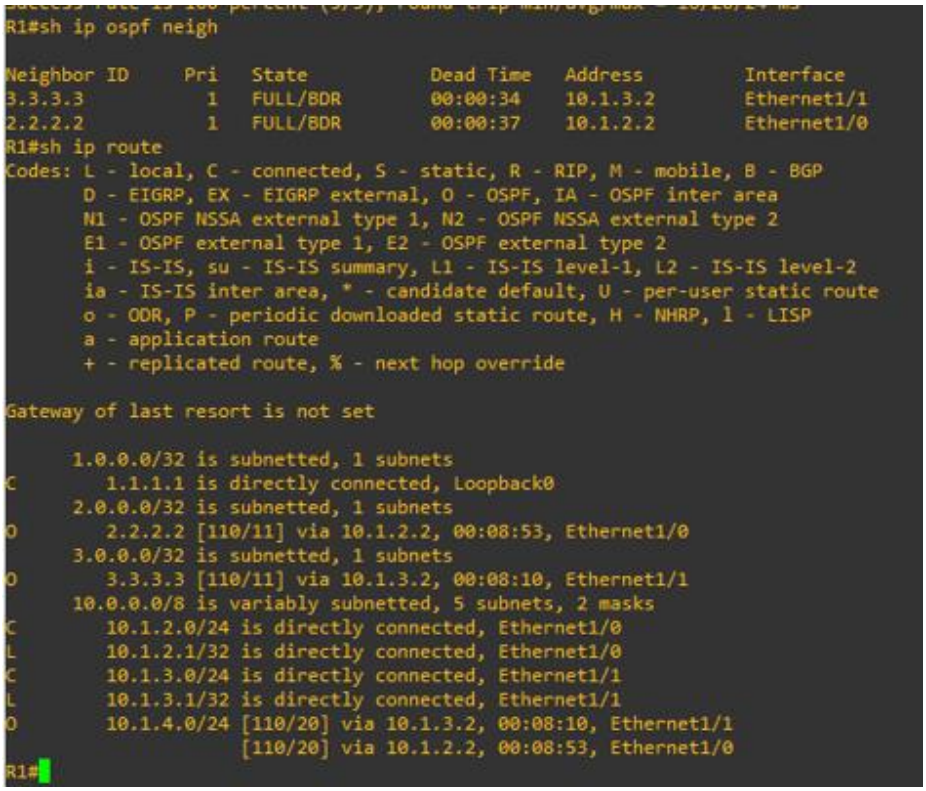


Για τον δρομολογητή R3:



5) Αρχικά με το command sh ip ospf neigh βλέπουμε ότι πράγματι υπάρχει η σύνδεση μεταξύ των δρομολογητών και στην συνέχεια παρατηρούμε με το command sh ip route βλέπουμε όλες τις συνδέσεις που έχουμε κάνει στους δρομολογητές.

Για το R1:



Για το R2:

```
R2#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:36    10.1.4.2        Ethernet1/1
1.1.1.1           1   FULL/DR         00:00:36    10.1.2.1        Ethernet1/0
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/11] via 10.1.2.1, 00:09:10, Ethernet1/0
      2.0.0.0/32 is subnetted, 1 subnets
C        2.2.2.2 is directly connected, Loopback0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/11] via 10.1.4.2, 00:08:26, Ethernet1/1
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C        10.1.2.0/24 is directly connected, Ethernet1/0
L        10.1.2.2/32 is directly connected, Ethernet1/0
O        10.1.3.0/24 [110/20] via 10.1.4.2, 00:08:26, Ethernet1/1
                     [110/20] via 10.1.2.1, 00:09:10, Ethernet1/0
C        10.1.4.0/24 is directly connected, Ethernet1/1
L        10.1.4.1/32 is directly connected, Ethernet1/1
R2#
```

Για το R3:

```
R3#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:31    10.1.4.1        Ethernet1/1
1.1.1.1           1   FULL/DR         00:00:31    10.1.3.1        Ethernet1/0
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/11] via 10.1.3.1, 00:07:07, Ethernet1/0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/11] via 10.1.4.1, 00:07:07, Ethernet1/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O        10.1.2.0/24 [110/20] via 10.1.4.1, 00:07:07, Ethernet1/1
                     [110/20] via 10.1.3.1, 00:07:07, Ethernet1/0
C        10.1.3.0/24 is directly connected, Ethernet1/0
L        10.1.3.2/32 is directly connected, Ethernet1/0
C        10.1.4.0/24 is directly connected, Ethernet1/1
L        10.1.4.2/32 is directly connected, Ethernet1/1
R3#
```

2ο Στάδιο Εργασίας

Στο δεύτερο στάδιο της εργασίας θα εκτελέσουμε την εντολή ping για να διαπιστώσουμε ότι υπάρχει η σύνδεση μεταξύ των router. Που στην συνέχεια θα εκτελέσουμε το command traceroute με αποτέλεσμα να δούμε τις διαδρομές τις δικτύωσης αλλά και τη μέτρηση των καθυστερήσεων μεταφοράς πακέτων για το πρωτοκόλλου Διαδικτύου.

Ερωτήματα

1) Πρώτα για να μπορούμε να κάνουμε σωστά το command ping 8.8.8.8 θα χρειαστεί να εκτελέσουμε σωστά τις εντολές και τα βήματα του link όπως να ορίσουμε στο IP του 8.8.8.8 την διεύθυνση google.com με αποτέλεσμα να έχουμε:

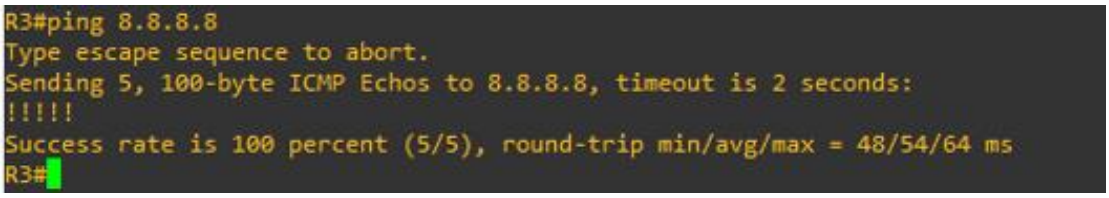
Για το R1:



Για το R2:



Για το R3:



2) Κάνοντας traceroute βλέπουμε την διαδρομή που ακολουθεί και ότι υπάρχει η σύνδεση με αποτέλεσμα να έχουμε:

```
R2#traceroute 8.8.8.8
Type escape sequence to abort.
Tracing the route to dns.google (8.8.8.8)
VRF info: (vrf in name/id, vrf out name/id)
  1 10.1.2.1 12 msec 20 msec 24 msec
  2 192.168.171.2 24 msec 28 msec 28 msec
  3  *  *  *
  4  *  *  *
  5  *  *  *
  6  *  *  *
  7  *  *  *
  8  *  *  *
  9  *  *  *
 10  *  *  *
 11  *  *  *
 12  *  *  *
 13  *  *  *
 14  *  *  *
 15  *  *  *
 16  *  *  *
 17  *  *  *
 18  *  *  *
 19  *  *  *
 20  *  *  *
 21  *  *  *
 22  *  *  *
 23  *  *  *
 24  *  *  *
 25  *  *  *
 26  *  *  *
 27  *  *  *
 28  *  *  *
 29  *  *  *
 30  *  *  *
R2#
```